

Scheme 5

Psychological Concept Clustering and Framework Mapping Scheme

Second-order normalization of detected psychological concepts

DRAFT FOR EXPERT EVALUATION. These coding schemes are a working draft circulated for expert evaluation. They have not been applied to a final review corpus. The current manuscript is a test run that exercised an earlier, coarser generation of these schemes in the Python workflow with an LLM (gemini-2.5-flash); the full end-to-end run is deliberately held until after expert feedback, and the refinements proposed here are awaiting sign-off before any re-run.

At a Glance

Workflow position

Post-detection concept normalization over Stage 2 and Stage 3 concept strings.

Operational mode

Fixed pattern-based concept detection upstream, followed by LLM clustering for higher-order normalization.

Unit of analysis

The set of unique concept strings across the corpus, not whole records.

Provenance basis

coding.py concept and framework patterns plus the clustering prompt.

Research questions

RQ3 (which psychological concepts and frameworks dominate)

Tagline

Fixed pattern detection upstream, LLM clustering for higher-order normalization

Canonical Source Paths

- `src/bps_review/extraction/coding.py`
- `src/protocol/codebooks/stage2_codebook.md`
- `src/protocol/codebooks/stage3_codebook.md`
- `src/data/interim/extraction/llm_concept_clusters.json`

Purpose

This scheme standardizes higher-order concept mapping after concept detection. It groups extracted psychological concepts from chronic pain review records into interpretable families and links them to likely theoretical frameworks. The goal is cross-record comparability when the raw concepts are heterogeneous, overlapping, or variably named.

It is a second-order coding scheme: it does not classify whole records, it normalizes the concept vocabulary itself.

Upstream Concept Detection Seeds (current)

The 16 fixed patterns currently used to detect concepts. The proposed taxonomy below expands and structures these.

- fear-avoidance, catastrophizing, kinesiophobia, depression, anxiety, coping, self-efficacy, acceptance
- illness perceptions, pain beliefs, distress, expectations, emotion regulation, mindfulness, trauma, sleep

Upstream Framework Seeds (current)

The 7 fixed framework patterns.

- fear-avoidance model, cognitive behavioral therapy, acceptance and commitment therapy
- illness perception framework, operant learning, social cognitive theory, self-regulation

Cluster Output Schema (current)

The clustering LLM returns strict JSON with a top-level clusters list.

clusters

Top-level list of normalized concept clusters.

family

Higher-order concept family label for the cluster.

members

The original concept strings grouped into that family. *(free text)*

possible_frameworks

Likely theoretical or therapeutic frameworks associated with the family. *(free text)*

Proposed Refinement: Ontology-Aligned Lexicon and Typed Relations

PROPOSED REFINEMENT

Awaiting expert evaluation. Not yet applied to the pipeline.

Refinement proposed for expert evaluation. The upstream detector recognizes only 16 concepts, which caps the resolution of RQ3 and misses constructs that appear in the corpus (for example resilience, suffering, functioning, maladaptive beliefs, pain-related thoughts). We propose expanding the detection lexicon and aligning every concept family to one of the 20 psychological subdomains defined in Scheme 6, so the concept map and the semantic ontology share one vocabulary.

We propose enriching the cluster schema so each cluster also carries parent_subdomain (a Scheme 6 label), definitional_status aggregated from Stage 3, and typed relations between families (is-a, part-of, associative). Members would each carry the verbatim source label alongside the canonical name.

Finally we propose two validation rules: every detected concept must map to exactly one family (no orphans), and every family must name at least one candidate framework or be explicitly marked framework-free. These rules make the clustering auditable and reproducible rather than free-form.

Comprehensive Psychological Concept Taxonomy

PROPOSED REFINEMENT

Awaiting expert evaluation. Not yet applied to the pipeline.

Refinement proposed for expert evaluation. This is the comprehensive concept space that the clustering step should normalize toward. Each family is aligned one to one with a psychological subdomain of the Scheme 6 ontology, so the concept map and the semantic ontology share a single vocabulary. Members are representative constructs (not exhaustive); frameworks are the theories most often invoked for that family in chronic pain research. Experts are asked to add, move, rename, or remove members and frameworks.

Catastrophizing and negative cognitive appraisal

Exaggerated negative interpretation and mental amplification of pain and its threat value. *Ontology subdomain:* Catastrophizing and Negative Cognitive Appraisal. *Constructs:* pain catastrophizing, rumination, magnification, helplessness, negative appraisal, worry about pain. *Frameworks:* fear-avoidance model, cognitive-behavioral model, communal coping model.

Fear, avoidance and pain-related fear

Fear of pain, movement, or reinjury and the avoidance or escape behaviour it motivates. *Ontology subdomain:* Fear Avoidance and Pain Related Fear. *Constructs:* kinesiophobia, fear of movement or reinjury, pain-related fear, avoidance behaviour, escape behaviour, threat hypervigilance. *Frameworks:* fear-avoidance model, avoidance-endurance model, operant learning.

Depression, low mood and negative affect

Depressive symptoms and sustained negative affect linked to living with pain. *Ontology subdomain:* Depression Emotional Distress and Affect. *Constructs:* depression, depressive symptoms, anhedonia, hopelessness, demoralization, negative affect. *Frameworks:* cognitive model of depression, diathesis-stress, mutual maintenance model.

Anxiety and psychological reactivity

Anxious apprehension and heightened physiological or cognitive reactivity to pain. *Ontology subdomain:* Anxiety and Psychological Reactivity. *Constructs:* anxiety, pain anxiety, health anxiety, anxiety sensitivity, physiological reactivity. *Frameworks:* anxiety sensitivity model, fear-avoidance model.

Self-efficacy, control and mastery

Beliefs about ability to manage pain and exert control over outcomes. *Ontology subdomain:* Self Efficacy Control Beliefs and Perceived Mastery. *Constructs:* pain self-efficacy, perceived control, locus of control, mastery, agency, confidence. *Frameworks:* social cognitive theory, self-efficacy theory.

Acceptance, psychological flexibility and mindfulness

Willingness to experience pain without avoidance while pursuing valued activity. *Ontology subdomain:* Acceptance Psychological Flexibility and Mindfulness. *Constructs:* pain acceptance, psychological flexibility, values-based action, committed action, present-moment awareness, cognitive defusion, willingness. *Frameworks:* acceptance and commitment therapy, psychological flexibility model, mindfulness-based approaches.

Coping strategies and adjustment

Cognitive and behavioural efforts to manage pain and the resulting adjustment. *Ontology subdomain:* Pain Coping Strategies and Adjustment. *Constructs:* active coping, passive coping, problem-focused coping, emotion-focused coping, adaptive coping, maladaptive coping, adjustment. *Frameworks:* transactional model of stress and coping, self-regulation.

Attention, vigilance and pain processing

How attention is captured by or directed away from pain and bodily threat. *Ontology subdomain:* Attention Vigilance and Pain Processing. *Constructs:* attentional bias, hypervigilance, distraction, pain

interruption, attentional control, somatic focus. *Frameworks*: threat interpretation model, cognitive-affective model of interruption.

Illness beliefs, pain representations and meaning

Beliefs about the nature, cause, timeline, and consequences of pain and its meaning. *Ontology subdomain*: Illness Beliefs Pain Representations and Meaning. *Constructs*: illness perceptions, pain beliefs, causal attributions, timeline beliefs, consequence beliefs, meaning of pain, illness identity. *Frameworks*: common-sense model of self-regulation, illness perception framework.

Cognitive-behavioural and psychotherapeutic approaches

Structured psychological techniques targeting pain cognition and behaviour. *Ontology subdomain*: Cognitive Behavioral and Psychotherapeutic Approaches. *Constructs*: cognitive restructuring, behavioural activation, graded activity, graded exposure, relaxation, psychoeducation. *Frameworks*: cognitive-behavioral therapy, operant and respondent conditioning.

Third-wave, ACT and contextual approaches

Contextual and acceptance-based therapies that target the function of experience. *Ontology subdomain*: Third Wave Therapies ACT and Contextual Approaches. *Constructs*: acceptance, defusion, values clarification, self-as-context, mindfulness practice, functional contextualism. *Frameworks*: acceptance and commitment therapy, contextual behavioural science, mindfulness-based stress reduction.

Resilience, positive psychology and growth

Protective strengths and positive adaptation despite persistent pain. *Ontology subdomain*: Resilience Positive Psychology and Post Traumatic Growth. *Constructs*: resilience, optimism, hope, benefit-finding, post-traumatic growth, positive affect, gratitude. *Frameworks*: broaden-and-build theory, resilience frameworks.

Identity, self-concept and pain biography

How pain reshapes self-concept, roles, and life narrative. *Ontology subdomain*: Identity Self Concept and Chronic Pain Biography. *Constructs*: pain identity, self-discrepancy, biographical disruption, role loss, self-concept, loss of self. *Frameworks*: self-discrepancy theory, narrative and biographical approaches.

Trauma, adversity and life events

Traumatic and adverse experiences that predispose to or maintain chronic pain. *Ontology subdomain*: Trauma Adverse Childhood and Life Events. *Constructs*: post-traumatic stress, trauma, adverse childhood experiences, abuse history, life stressors, victimization. *Frameworks*: mutual maintenance model, diathesis-stress, shared vulnerability model.

Personality and individual differences

Stable traits and dispositions that shape pain experience and response. *Ontology subdomain*: Personality Psychological Traits and Individual Differences. *Constructs*: neuroticism, negative affectivity, harm avoidance, perfectionism, alexithymia, trait anxiety. *Frameworks*: trait vulnerability models, diathesis-stress models.

Cognitive function and executive processes

Cognitive resources and executive processes affected by or affecting pain. *Ontology subdomain*: Cognitive Function Executive Processes and Brain Health. *Constructs*: attention, working memory, executive function, cognitive load, mental fatigue or brain fog, processing speed. *Frameworks*: cognitive resource models, interruption models.

Motivation, goal pursuit and engagement

Motivational dynamics of pursuing valued goals alongside pain. *Ontology subdomain*: Motivational

Processes Goal Pursuit and Engagement. *Constructs*: goal pursuit, goal conflict, approach and avoidance motivation, activity engagement, endurance, valued goals. *Frameworks*: self-regulation, motivational control theory, goal pursuit models.

Healthcare-seeking, adherence and engagement

Behaviour around seeking, engaging with, and adhering to care. *Ontology subdomain*: Healthcare Seeking Treatment Adherence and Engagement. *Constructs*: treatment adherence, healthcare utilization, help-seeking, therapeutic alliance, treatment expectations, engagement. *Frameworks*: health belief model, common-sense model of self-regulation, expectancy models.

Emotion regulation and pain affect processing

Strategies for regulating emotion and processing the affective dimension of pain. *Ontology subdomain*: Emotional Regulation and Pain Affect Processing. *Constructs*: emotion regulation, expressive suppression, cognitive reappraisal, alexithymia, emotional awareness, affect regulation. *Frameworks*: process model of emotion regulation, emotion regulation frameworks.

Mental health comorbidity and wellbeing

Co-occurring mental health conditions and overall psychological wellbeing. *Ontology subdomain*: Mental Health Comorbidity and Psychological Wellbeing. *Constructs*: comorbid depression or anxiety, distress, quality of life, wellbeing, suicidality, life satisfaction. *Frameworks*: mutual maintenance model, biopsychosocial wellbeing models.

Analytic Role

- Preserves the original detected concepts while improving comparability across records.
- Supports synthesis questions about which psychological concepts and frameworks dominate the corpus.
- Allows concept families to be interpreted above raw string matching.

Primary Outputs Using This Scheme

- `src/data/interim/extraction/llm_concept_clusters.json`